

SAHAKAR SETU — Idea Approach (Detailed)

Multilingual Cooperative Governance & Legal Assistance — PS 26088 (Ministry of Cooperation / NCCT)

Team: LexNova · Team ID 131559 · Smart India Hackathon 2026 · Theme: Agriculture, FoodTech & Rural Development · Mode: Software + Hardware (proposed)

This document explains, step by step, how a member's question becomes a **verified, cited answer** — and how a member's problem becomes a **tracked grievance** — across all four channels (IVR call, WhatsApp, Web/kiosk, SMS) that share one brain.

Step 1: Query Intake — one question, four doors

The member can reach Sahakar Setu through four channels. Every channel feeds the same pipeline, so a case opened on a phone call is visible on the web and continues on WhatsApp.

Channel	How it works	Internet needed?
IVR voice call (+13469986840)	Member calls a toll-free-style number. Vapi.ai handles the telephony; <code>talkscribe</code> (Whisper, language <code>hi</code>) transcribes speech; our server is the custom LLM (OpenAI-compatible endpoint <code>POST /webhook/vapi/chat/completions</code> , supports SSE streaming). Answer is spoken back by Vapi voice <code>Naina</code> (Hindi).	No — plain voice plan on a feature phone
Web / PWA / Kiosk	React PWA (also wrapped in Capacitor for the kiosk APK) with chat UI, voice capture (AudioWorklet) and playback. Kiosk adds lockdown, auto-start and thermal printing.	Yes
WhatsApp	Twilio webhook (<code>POST /webhook/twilio/whatsapp</code>) — same RAG + LLM pipeline, text in / text out.	Yes
SMS	Twilio webhook (<code>POST /webhook/twilio/sms</code>) — short answer by SMS; also the last-resort fallback when TTS fails on voice channels.	No

Language auto-detection

Every incoming query passes through the language detector (`src/modules/language/detector.ts`). Hindi, Hinglish (Roman-script Hindi) and English are handled natively; the detected language is stored on the session and drives both the answer language and the TTS voice.

Nothing is forgotten

Every turn (user + assistant) is persisted to the Supabase `messages` table (`storeTurn`), and every conversation is a `sessions` row (VAPI calls are keyed by `external_ref = 'vapi:<callId>'`). Recent history (last 4–6 turns) is injected into the next prompt, so the assistant remembers context across turns and channels.

Step 2: Retrieval — finding the law, not guessing it

Answers are grounded in a curated knowledge corpus in Supabase (`corpus_passages`): **333 passages** authored from cooperative acts/by-laws, government scheme circulars (PMFBY, PM-Kisan, KCC), PACS service manuals and financial-literacy material. Retrieval is a deterministic, explainable BM25 keyword search (no embeddings dependency in the critical path):

Mechanism	Value	Why
Hinglish keyword expansion (<code>HINGLISH_MAP</code>)	e.g. <code>member</code> → <code>sadasya</code> , <code>sadasyata</code> ; <code>premium</code> → <code>kiraya</code> , <code>bhugtan</code>	Members mix Hindi words into Roman-script queries
Hindi stop-word removal	~70 words (<code>mai</code> , <code>kya</code> , <code>hai</code> , <code>ka</code> , <code>ki</code> ...)	Keeps scoring focused on content words
BM25 term scoring	<code>k1 = 1.2</code> , <code>b = 0.75</code> , <code>IDF = log(1 + (N - df + 0.5)/(df + 0.5))</code>	Standard Okapi BM25 — length-normalized term frequency
Bigram boost	<code>+3.0</code> per exact bigram match	Rewards phrase matches ("crop insurance" vs "crop" + "insurance")
Full-phrase boost	<code>+2.5</code> when the whole query (<code>len > 8</code>) appears in the passage	Exact question phrasing is a strong signal
Threshold + top-K	<code>score ≥ 0.2</code> , top 5	Cuts noise; keeps the prompt small for low latency
Deterministic order	ties broken by <code>ORDER BY id</code>	Same query ⇒ same passages ⇒ reproducible answers

Verified result: 13/13 guide queries return the correct source passage in the top-3, and the premium passage ranks #1 for "PMFBY premium kya hai?".

Step 3: Guardrailed Generation — answers that stay inside the fence

The retrieved passages are packaged with a strict system prompt (`src/modules/llm/groq.ts`) and sent to Groq:

Item	Value
Primary model	<code>openai/gpt-oss-120b</code> (text channels)
Voice-first model	<code>openai/gpt-oss-20b</code> (IVR/kiosk — lower latency)
Fallback	On any failure the other model is tried automatically (<code>fallbackTotal</code> metric)
Temperature	0.3 text / 0.7 voice (natural spoken variation)
Passage truncation	700 chars per passage (text), 250 (voice) — keeps prompt under Groq TPM caps

Prompt rules	Answer only from retrieved passages for legal/scheme facts; natural spoken Hindi; one follow-up question when a detail is missing; empathy in own words; no markdown on voice
Context bundle	Case facts ([FACT] memory), last 4–6 turns of history, passages with [doc, section] labels

Step 4: Citation Enforcement — no source, no answer

After generation, the answer is post-processed (`postProcessAnswer` + `extractCitations`):

- **Grounding retry:** on text channels, if passages were retrieved but the answer contains no [Source: ...] citation, the LLM is asked once more with an explicit citation-enforcement message (`groundingRetriesTotal` metric). If the retry yields citations, it replaces the answer.
- **Decline:** if grounding still yields no citation, the answer is replaced with the exact verified-decline phrase: “*Mere paas iska verified jawab nahi hai.*” — Sahakar Setu never presents an unverified legal claim as fact.
- **Voice cleanup:** on voice channels, citation tags and markdown are stripped before TTS, so the caller hears natural speech.

Step 5: The Honest Boundary — say what you can't do

Out-of-domain questions (recipes, cricket, movies, programming...) get exactly one verbatim response, enforced both in the prompt and by output canonicalization (`isBoundaryLike` normalizes Devanagari word variants like सिर्फ → sirf, I → . and requires ≥ 85% word overlap before re-canonicalizing to the exact boundary message):

“Main Sahakar Setu hoon aur sirf sahkari kanoon, sarkari yojanaon, PACS sevaon, PMFBY, vittiya saksharta aur shikayat samadhan se judi jaankari de sakta hoon. Aap in vishayon mein kuch pooch sakte hain.”

Step 6: From answer to action — facts, grievance, escalation

- **Fact memory:** after each turn, a lightweight extractor pulls structured facts (name, village, district, crop, amount, membership no...) from the conversation and saves them to the case (`case_facts` table, camelCase in code / snake_case in DB). These facts are re-injected into later prompts, so the member never repeats themselves.
- **Grievance drafting:** when a problem needs a written application, a Hindi (or English) Handlebars template fills in the facts and generates a formal application addressed to the right authority (8 categories: PACS, bank, PMFBY/insurance, Registrar, election, loan, subsidy, land). Each draft gets a unique tracking ID (format `SS-2026-XXXXXX`) and is stored in `grievance_drafts` + `grievance_tracking` .
- **Tracking & escalation:** the member can track the ID on any channel. A time-based escalation ladder computes the next step: **Society/PACS (15 days)** → **Registrar** → **Ombudsman (PMFBY, 30 days)** → **Legal Aid (NALSA, 60 days)**, with a ready draft template for each level.
- **Case strength (verdict):** a 0–100 score with the missing evidence listed — an honest verdict engine, no false hope. Cases scoring 60+ are offered verified-advocate connect.
- **Kiosk printout:** on the kiosk, the grievance draft and tracking receipt print on a thermal printer.

Graceful degradation — the service never goes silent

Failure	Fallback
Primary LLM error	Secondary model (120b ⇌ 20b)
TTS provider fails (e.g. edge-tts blocked on Render IPs)	gTTS fallback → voice answer still plays
All voice synthesis fails	Answer delivered by SMS
STT transcript empty	Polite re-ask: “Mujhe aapki awaaz sunai nahi di. Kripya dohrayein.”
Vosk worker crashes	Auto-respawn on next request (worker alive-check)

Why this works for rural India

- **Voice-first:** 2 of 4 channels need no internet; feature-phone compatible.
- **No hallucinations on legal facts:** citation-enforced, decline-and-escalate.
- **One case record** across call / WhatsApp / web / SMS.
- **Hinglish native:** keyword expansion + language detection handle code-mixing.
- **Affordable:** no new proprietary hardware — the kiosk is a phone + OTG hub + printer (see Kiosk Hardware doc).